

GATE DA (Data Science & AI) Practice Question Bank - Set 5

7 Topics · Previous-Year-Style + New Practice Questions · Full Solutions

This set covers seven topics: Expectation, Variance & Moments, Logistic Regression (Sigmoid & Decision Boundary), Relational Calculus (Tuple & Domain), Time Complexity, Nested Loop Complexity, the Master Theorem (5 recurrence problems), and Uninformed Search -- Uniform Cost Search (UCS).

SAME HONESTY POLICY AS BEFORE: GATE DA has run only in 2024, 2025 and 2026, and it is a non-coding, MCQ/MSQ/NAT written exam. Every "old" question below is tagged with its real provenance:

[VERIFIED PYQ] -- a genuine, well-documented past GATE question (year and paper stated). Where the source is GATE CSE rather than GATE DA specifically, that is stated explicitly.

[GATE-style] -- not a verbatim past question (either no verified PYQ exists on that exact sub-point, or the original figure/data could not be reliably reproduced), but written to match real GATE DA difficulty, phrasing, and mark-weightage.

Section B ("New Practice Questions") in every topic is freshly written GATE-level practice -- none of these are claimed to be past questions.

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Topic 1: Expectation, Variance & Moments

Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE DA 2025, Question 11]

Q1. [MCQ · 1 mark]

Suppose X and Y are random variables. The conditional expectation of X given Y is denoted $E[X|Y]$. Then $E[E[X|Y]]$ equals:

- (A) $E[XY]$
- (B) $E[Y]/E[X]$
- (C) $E[X]$
- (D) $E[Y]$

Answer: (C)

Solution: This is the Law of Total Expectation (law of iterated expectations): $E[E[X|Y]] = E[X]$, for any random variables X and Y .

[VERIFIED PYQ — GATE DA 2024]

Q2. [NAT · 2 marks]

X is an exponentially distributed random variable with rate λ . It is given that $5 \cdot E(X) = \text{Var}(X)$. Find the value of λ (correct to 1 decimal).

Answer: 0.2

Solution: For $\text{Exponential}(\lambda)$: $E(X) = 1/\lambda$, $\text{Var}(X) = 1/\lambda^2$. So $5(1/\lambda) = 1/\lambda^2 \Rightarrow 5\lambda = 1 \Rightarrow \lambda = 0.2$.

[GATE-style]

Q3. [NAT · 1 mark]

A discrete RV X has PMF $P(X=1)=0.2$, $P(X=2)=0.5$, $P(X=3)=0.3$. Find $E[X]$.

Answer: 2.1

Solution: $E[X] = 1(0.2) + 2(0.5) + 3(0.3) = 0.2 + 1.0 + 0.9 = 2.1$.

[GATE-style]

Q4. [NAT · 1 mark]

Using the PMF from the previous question, find $E[X^2]$.

Answer: 4.9

Solution: $E[X^2] = 1(0.2) + 4(0.5) + 9(0.3) = 0.2 + 2.0 + 2.7 = 4.9$.

[GATE-style]

Q5. [NAT · 2 marks]

Using the results of the previous two questions, find $\text{Var}(X)$.

Answer: 0.49

Solution: $\text{Var}(X) = E[X^2] - (E[X])^2 = 4.9 - (2.1)^2 = 4.9 - 4.41 = 0.49$.

[GATE-style]

Q6. [MCQ · 1 mark]

For any random variable X and constants a , b , $\text{Var}(aX+b)$ equals:

- (A) $a \cdot \text{Var}(X) + b$
- (B) $a^2 \text{Var}(X)$

- (C) $a^2\text{Var}(X)+b$
(D) $a\cdot\text{Var}(X)$

Answer: (B)

Solution: Adding a constant b does not change spread; scaling by a scales variance by a^2 .

[GATE-style]

Q7. [MCQ · 1 mark]

Variance relates to the second moment about the origin, $E[X^2]$, via:

- (A) $\text{Var}(X) = E[X^2] + (E[X])^2$
(B) $\text{Var}(X) = E[X^2] - (E[X])^2$
(C) $\text{Var}(X) = (E[X^2])^2 - E[X]$
(D) $\text{Var}(X) = E[X^2] / (E[X])^2$

Answer: (B)

[GATE-style]

Q8. [MSQ · 2 marks]

Which of the following are TRUE about expectation?

- (A) $E[X+Y] = E[X]+E[Y]$, always, regardless of independence (linearity)
(B) $E[XY] = E[X]E[Y]$, always, regardless of independence
(C) $E[aX+b] = aE[X]+b$
(D) If X and Y are independent, then $E[XY] = E[X]E[Y]$

Answer: (A), (C), (D)

Solution: (B) is false in general -- $E[XY]=E[X]E[Y]$ only holds under independence (or more generally, uncorrelatedness); it is not a universal property of expectation.

[GATE-style]

Q9. [NAT · 1 mark]

For $X \sim \text{Binomial}(n=10, p=0.4)$, find $E[X]$.

Answer: 4

Solution: $E[X] = np = 10 \times 0.4 = 4$.

[GATE-style]

Q10. [NAT · 1 mark]

For $X \sim \text{Binomial}(n=10, p=0.4)$ (as in the previous question), find $\text{Var}(X)$.

Answer: 2.4

Solution: $\text{Var}(X) = np(1-p) = 10 \times 0.4 \times 0.6 = 2.4$.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

A discrete RV X has PMF $P(X=0)=0.1$, $P(X=1)=0.4$, $P(X=2)=0.5$. Find $E[X]$.

Answer: 1.4

Solution: $E[X] = 0(0.1)+1(0.4)+2(0.5) = 0.4+1.0 = 1.4$.

[New Practice]

Q12. [NAT · 2 marks]

Using the PMF from the previous question, find $\text{Var}(X)$.

Answer: 0.44

Solution: $E[X^2] = 0(0.1) + 1(0.4) + 4(0.5) = 0.4 + 2.0 = 2.4$. $\text{Var}(X) = 2.4 - (1.4)^2 = 2.4 - 1.96 = 0.44$.

[New Practice]

Q13. [MCQ · 1 mark]

The third central moment of a distribution is primarily used to measure:

- (A) Central tendency
- (B) Skewness (asymmetry)
- (C) Spread/variance
- (D) Kurtosis

Answer: (B)

[New Practice]

Q14. [MCQ · 1 mark]

The fourth standardized central moment of a distribution is primarily used to measure:

- (A) Mean
- (B) Variance
- (C) Skewness
- (D) Kurtosis (peakedness / tail heaviness)

Answer: (D)

[New Practice]

Q15. [MSQ · 2 marks]

Which of the following are TRUE regarding the covariance of two random variables X and Y?

- (A) $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$
- (B) If X and Y are independent, $\text{Cov}(X, Y) = 0$
- (C) $\text{Cov}(X, Y) = 0$ always implies X and Y are independent
- (D) $\text{Cov}(X, X) = \text{Var}(X)$

Answer: (A), (B), (D)

Solution: (C) is false -- zero covariance means no LINEAR relationship, but X and Y can still be dependent through a non-linear relationship (e.g. $Y = X^2$ for symmetric X can have zero covariance yet clear dependence).

[New Practice]

Q16. [NAT · 1 mark]

For $X \sim \text{Poisson}(\lambda=3)$, both $E[X]$ and $\text{Var}(X)$ equal λ . Find this common value.

Answer: 3

[New Practice]

Q17. [MCQ · 1 mark]

For a standard normal random variable Z, $E[Z]$ and $\text{Var}(Z)$ are respectively:

- (A) 0 and 1
- (B) 1 and 0

- (C) 0 and 0
- (D) 1 and 1

Answer: (A)

[New Practice]

Q18. [NAT · 1 mark]

X and Y are independent with $\text{Var}(X)=4$ and $\text{Var}(Y)=9$. Find $\text{Var}(X+Y)$.

Answer: 13

Solution: For independent RVs, $\text{Var}(X+Y) = \text{Var}(X)+\text{Var}(Y) = 4+9 = 13$.

[New Practice]

Q19. [NAT · 1 mark]

Using the same X, Y as the previous question (independent, $\text{Var}(X)=4$, $\text{Var}(Y)=9$), find $\text{Var}(X-Y)$.

Answer: 13

Solution: For independent RVs, variance still ADDS for the difference: $\text{Var}(X-Y) = \text{Var}(X)+\text{Var}(Y) = 13$ (the minus sign does not subtract variances).

[New Practice]

Q20. [MCQ · 1 mark]

The standard deviation of a random variable X is defined as:

- (A) $\text{Var}(X)$
- (B) $\sqrt{\text{Var}(X)}$
- (C) $E[X]$
- (D) $E[X^2]$

Answer: (B)

Topic 2: Logistic Regression - Sigmoid & Decision Boundary

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [MCQ · 1 mark]

The sigmoid (logistic) function $\sigma(z)$ is defined as:

- (A) $\sigma(z) = 1/(1+e^{-z})$
- (B) $\sigma(z) = e^z/(1+z)$
- (C) $\sigma(z) = z/(1+z^2)$
- (D) $\sigma(z) = 1/(1-e^{-z})$

Answer: (A)

[GATE-style]

Q2. [NAT · 1 mark]

Compute $\sigma(0)$, the value of the sigmoid function at $z=0$.

Answer: 0.5

Solution: $\sigma(0) = 1/(1+e^0) = 1/(1+1) = 0.5$.

[GATE-style]

Q3. [MCQ · 1 mark]

As $z \rightarrow +\infty$, $\sigma(z)$ approaches:

- (A) 0
- (B) 0.5
- (C) 1
- (D) ∞

Answer: (C)

[GATE-style]

Q4. [MCQ · 1 mark]

As $z \rightarrow -\infty$, $\sigma(z)$ approaches:

- (A) 0
- (B) 0.5
- (C) 1
- (D) $-\infty$

Answer: (A)

[GATE-style]

Q5. [MCQ · 1 mark]

In logistic regression, the decision boundary (for a classification threshold of 0.5) is defined by:

- (A) $w^T x + b = 0.5$
- (B) $w^T x + b = 0$
- (C) $w^T x + b = 1$
- (D) $\sigma(w^T x + b) = 1$

Answer: (B)

Solution: $\sigma(z)=0.5$ exactly when $z=0$, so the decision boundary is the set of points where $w^T x + b = 0$.

[GATE-style]

Q6. [MCQ · 1 mark]

The decision boundary produced by standard logistic regression (no polynomial/interaction features), in terms of the original input features, is always:

- (A) A circle
- (B) Linear (a straight line / hyperplane)
- (C) A parabola
- (D) Always non-linear

Answer: (B)

[GATE-style]

Q7. [NAT · 2 marks]

A logistic regression model has weights $w=[2,-1]$ and bias $b=-1$ (two features x_1, x_2). Find $z = w^T x + b$ for the point $x=(1,1)$.

Answer: 0

Solution: $z = 2(1) + (-1)(1) + (-1) = 2 - 1 - 1 = 0$ (the point lies exactly on the decision boundary).

[GATE-style]

Q8. [MCQ · 1 mark]

For the point in the previous question ($z=0$ exactly), the predicted probability $\sigma(z)$ is:

- (A) 0
- (B) 0.5
- (C) 1
- (D) Undefined

Answer: (B)

[GATE-style]

Q9. [MSQ · 2 marks]

Which of the following are TRUE about the sigmoid function used in logistic regression?

- (A) Its output always lies strictly between 0 and 1
- (B) It is monotonically increasing
- (C) Its derivative can be written as $\sigma(z)(1-\sigma(z))$
- (D) It maps large negative inputs to values close to 1

Answer: (A), (B), (C)

Solution: (D) is false -- large negative z maps close to 0, not 1.

[GATE-style]

Q10. [MCQ · 1 mark]

To convert the sigmoid's output probability into a hard binary class prediction using a 0.5 threshold, predict class 1 whenever:

- (A) $\sigma(z) \geq 0.5$, equivalently $z \geq 0$
- (B) $\sigma(z) \leq 0.5$
- (C) $z \leq 0$

(D) $\sigma(z) = 0$ exactly

Answer: (A)

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

For logistic regression weights $w=[1,2]$, bias $b=-5$, find $z=w^T x+b$ for $x=(2,1)$.

Answer: -1

Solution: $z = 1(2)+2(1)-5 = 2+2-5 = -1$.

[New Practice]

Q12. [MCQ · 1 mark]

Since $z=-1$ (negative) for the point in the previous question, the predicted class (0.5 threshold) is:

- (A) Class 0
- (B) Class 1
- (C) Cannot be determined
- (D) Exactly on the boundary

Answer: (A)

Solution: $z < 0$ means $\sigma(z) < 0.5$, so the predicted class is 0.

[New Practice]

Q13. [NAT · 2 marks]

Compute $\sigma(2)$, the sigmoid function value at $z=2$ (correct to 4 decimals).

Answer: 0.8808

Solution: $\sigma(2) = 1/(1+e^{-2}) = 1/(1+0.1353) = 1/1.1353 \approx 0.8808$.

[New Practice]

Q14. [MCQ · 1 mark]

The logistic regression cost function (log-loss) for a single training example is:

- (A) $-[y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$
- (B) $(y-\hat{y})^2$
- (C) $|y-\hat{y}|$
- (D) $y \cdot \hat{y}$

Answer: (A)

[New Practice]

Q15. [MSQ · 2 marks]

Which of the following are TRUE about logistic regression?

- (A) It models the log-odds (logit) of the positive class as a linear function of the input features
- (B) It directly predicts continuous, unrestricted real-valued outputs
- (C) Its cost function (log-loss / cross-entropy) is convex, guaranteeing a unique global minimum
- (D) It can be extended to multi-class classification via one-vs-rest or softmax

Answer: (A), (C), (D)

Solution: (B) is false -- logistic regression's output is a probability strictly confined to (0,1).

[New Practice]

Q16. [NAT · 1 mark]

For weights $w=[3]$ and bias $b=-6$ (single feature x), find the value of x at which the decision boundary occurs (where $z=0$).

Answer: 2

Solution: $z = 3x - 6 = 0 \Rightarrow x = 2$.

[New Practice]

Q17. [MCQ · 2 marks]

For the decision boundary $w_1x_1 + w_2x_2 + b = 0$ with $w_1=1$, $w_2=1$, $b=-4$, which of the following points lies exactly ON the decision boundary?

- (A) (1,1)
- (B) (2,2)
- (C) (0,0)
- (D) (5,5)

Answer: (B)

Solution: (2,2): $2+2-4 = 0$ ✓. The others do not satisfy $z=0$.

[New Practice]

Q18. [NAT · 1 mark]

Using $w=[1,1]$, $b=-4$ from the previous question, compute z for the point (3,3).

Answer: 2

Solution: $z = 3+3-4 = 2$.

[New Practice]

Q19. [MCQ · 1 mark]

Since $z=2$ (positive) for the point (3,3) above, this point is classified (0.5 threshold) as:

- (A) Class 0
- (B) Class 1
- (C) Exactly on the boundary
- (D) Cannot say

Answer: (B)

[New Practice]

Q20. [MCQ · 2 marks]

Logistic regression can be made to produce a non-linear (e.g. circular or elliptical) decision boundary by:

- (A) It's impossible -- logistic regression is always strictly linear
- (B) Adding polynomial/interaction features (e.g. x_1^2 , x_2^2 , x_1x_2) before fitting the linear model in the transformed feature space
- (C) Increasing the learning rate
- (D) Using a smaller regularization parameter

Answer: (B)

Topic 3: Relational Calculus - Tuple & Domain

Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE CSE 2006]

Q1. [MCQ · 2 marks]

Which of the following relational query languages have the same expressive power? I) Relational algebra. II) Tuple relational calculus restricted to safe expressions. III) Domain relational calculus restricted to safe expressions.

- (A) II and III only
- (B) I and II only
- (C) I and III only
- (D) I, II and III

Answer: (D)

Solution: This is Codd's theorem -- relational algebra, safe tuple relational calculus, and safe domain relational calculus are all equivalent in expressive power ("relationally complete").

[VERIFIED PYQ — GATE CSE 2014, Set 3, concept]

Q2. [MCQ · 1 mark]

Which of the following tuple relational calculus expressions is equivalent to $\forall t \in r (P(t))$?

- (A) $\neg \exists t \in r (\neg P(t))$
- (B) $\exists t \in r (\neg P(t))$
- (C) $\neg \exists t \in r (P(t))$
- (D) $\exists t \notin r (P(t))$

Answer: (A)

Solution: "For all tuples t in r , $P(t)$ holds" is logically equivalent to "there does not exist a tuple t in r for which $P(t)$ is false" -- a standard quantifier-negation identity.

[GATE-style]

Q3. [MCQ · 1 mark]

In Tuple Relational Calculus (TRC), a query has the general form $\{t \mid P(t)\}$, where t is:

- (A) A single attribute value
- (B) A tuple variable, ranging over tuples of some relation
- (C) An entire relation
- (D) A numeric constant

Answer: (B)

[GATE-style]

Q4. [MCQ · 1 mark]

In Domain Relational Calculus (DRC), a query has the general form $\{ \langle x_1, \dots, x_n \rangle \mid P(x_1, \dots, x_n) \}$, where each x_i is:

- (A) A tuple variable
- (B) A domain variable, ranging over values from an attribute's domain
- (C) A relation name
- (D) A Boolean flag

Answer: (B)

[GATE-style]

Q5. [MCQ · 1 mark]

The existential quantifier $\exists$ in relational calculus is used to express:

- (A) "For all tuples, condition P holds"
- (B) "There exists at least one tuple satisfying condition P"
- (C) "No tuple satisfies P"
- (D) "Exactly one tuple satisfies P"

Answer: (B)

[GATE-style]

Q6. [MCQ · 1 mark]

A relational calculus query is called "safe" if:

- (A) It uses only the AND operator
- (B) It is guaranteed to yield a finite result (avoiding infinite relations, e.g. from unrestricted negation or universal quantification)
- (C) It uses no quantifiers at all
- (D) It always returns exactly one tuple

Answer: (B)

[GATE-style]

Q7. [MCQ · 1 mark]

When translating between TRC selection predicates and relational algebra conditions, which relational calculus connectives correspond to logical AND, OR, and NOT respectively?

- (A) $\cup$, $\cap$, $-$
- (B) $\wedge$, $\vee$, $\neg$
- (C) $\forall$, $\exists$, $\neg\exists$
- (D) σ , π , ρ

Answer: (B)

[GATE-style]

Q8. [MSQ · 2 marks]

Which of the following are TRUE about Tuple Relational Calculus (TRC)?

- (A) It is a declarative (non-procedural) query language, unlike relational algebra which is procedural
- (B) TRC queries specify WHAT data to retrieve, not HOW to retrieve it
- (C) Every syntactically valid TRC expression is guaranteed to be "safe" (a finite result)
- (D) TRC and relational algebra (restricted to safe expressions) have the same expressive power

Answer: (A), (B), (D)

Solution: (C) is false -- unrestricted TRC expressions (e.g. using unbounded negation) can define infinite relations and are therefore "unsafe"; safety must be explicitly ensured.

[GATE-style]

Q9. [MCQ · 1 mark]

SQL's core SELECT-FROM-WHERE query structure is most closely based on which formal query language?

- (A) Relational Algebra
- (B) Tuple Relational Calculus
- (C) Domain Relational Calculus
- (D) None of the above

Answer: (B)

[GATE-style]

Q10. [MCQ · 1 mark]

The universal quantifier $\forall$ in relational calculus can always be rewritten using existential quantification and negation as:

- (A) $\forall t(P(t)) \equiv \exists t(\neg P(t))$
- (B) $\forall t(P(t)) \equiv \neg \exists t(\neg P(t))$
- (C) $\forall t(P(t)) \equiv \exists t(P(t))$
- (D) $\forall t(P(t)) \equiv \neg \forall t(\neg P(t))$

Answer: (B)

Section B — New Practice Questions (10)

[New Practice]

Q11. [MCQ · 2 marks]

For relation Employee(name, dept, salary), which TRC expression correctly retrieves the names of employees earning more than 50000?

- (A) $\{t.name \mid \text{Employee}(t) \wedge t.salary > 50000\}$
- (B) $\{t \mid t.name > 50000\}$
- (C) $\{t.name \mid t.salary\}$
- (D) $\pi_{name}(\sigma_{salary > 50000}(\text{Employee}))$

Answer: (A)

Solution: (D) is the relational algebra equivalent, not TRC notation -- (A) is the correct TRC form.

[New Practice]

Q12. [MCQ · 2 marks]

Which of the following is the DRC equivalent of "find names of employees earning more than 50000" for relation Employee(name, dept, salary)?

- (A) $\{ \langle n \rangle \mid \exists d, s (\langle n, d, s \rangle \in \text{Employee} \wedge s > 50000) \}$
- (B) $\{ \langle n, d, s \rangle \mid \text{Employee}(n, d, s) \}$
- (C) $\{ n \mid n > 50000 \}$
- (D) $\pi_{name}(\text{Employee})$

Answer: (A)

[New Practice]

Q13. [NAT · 1 mark]

Relation R(A,B,C) has 100 tuples. A TRC query $\{t.A \mid R(t) \wedge t.B = 5\}$ finds 15 matching tuples. How many tuples does the result relation contain (assuming all resulting A-values happen to be distinct)?

Answer: 15

[New Practice]

Q14. [MSQ · 2 marks]

Which of the following are TRUE about the relationship between relational algebra and relational calculus?

- (A) Every relational algebra expression can be converted into an equivalent safe TRC (or DRC) expression
- (B) Every safe TRC/DRC expression can be converted into an equivalent relational algebra expression
- (C) Relational calculus is procedural, specifying the exact sequence of operations
- (D) The equivalence of relational algebra and safe relational calculus is known as Codd's theorem / relational completeness

Answer: (A), (B), (D)

Solution: (C) is false -- relational algebra is procedural, while relational calculus is declarative (non-procedural).

[New Practice]

Q15. [MCQ · 2 marks]

In DRC, the expression $\{ \langle x, y \rangle \mid \exists z (R(x, z) \wedge S(z, y)) \}$ is most closely equivalent to which relational algebra operation between R and S?

- (A) Union
- (B) A join of R and S on their shared attribute (via z), projected onto x,y
- (C) Set difference
- (D) Cartesian product only, without any condition

Answer: (B)

[New Practice]

Q16. [MCQ · 1 mark]

Which of the following is a key structural difference between TRC and DRC?

- (A) TRC variables range over whole tuples, while DRC variables range over individual domain (attribute) values
- (B) TRC does not support quantifiers, while DRC does
- (C) DRC cannot express selection conditions
- (D) There is no meaningful difference between TRC and DRC

Answer: (A)

[New Practice]

Q17. [NAT · 1 mark]

A safe TRC query over relation R (200 tuples) and relation S (150 tuples) can, by the safety restriction, only return tuples built from values that actually appear in R or S. If the query's result is a subset of R's tuples only, what is the maximum possible number of tuples the result could contain?

Answer: 200

[New Practice]

Q18. [MCQ · 1 mark]

The expression $\{ t \mid \neg(t \in R) \}$ (all tuples NOT in R) is generally considered:

- (A) Safe, since R is a finite relation

- (B) Unsafe, because it could define an infinite set of tuples not in R (over an infinite domain of possible tuple values)
- (C) Always empty
- (D) Equivalent to R itself

Answer: (B)

[New Practice]

Q19. [MCQ · 1 mark]

Which of the following relational calculus features requires special care to ensure the "safety" (finiteness) of a query's result?

- (A) Use of AND ($\wedge$) between two conditions
- (B) Use of negation ($\neg$) and/or universal quantification ($\forall$), which can implicitly reference an unbounded domain
- (C) Simple attribute comparisons like $t.A = 5$
- (D) Projection of a single attribute

Answer: (B)

[New Practice]

Q20. [MCQ · 1 mark]

Codd's theorem states that a query language is "relationally complete" if it is at least as expressive as:

- (A) SQL
- (B) Basic relational algebra
- (C) The C programming language
- (D) First-order predicate logic without restriction

Answer: (B)

Topic 4: Time & Work

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [NAT · 1 mark]

A can complete a work in 10 days, and B can complete the same work in 15 days. If they work together, how many days will they take to complete the work?

Answer: 6

Solution: Rate of A = $1/10$, rate of B = $1/15$. Combined rate = $1/10 + 1/15 = 3/30 + 2/30 = 5/30 = 1/6$.
Time = 6 days.

[GATE-style]

Q2. [NAT · 1 mark]

A can do a piece of work in 12 days. In how many days will A complete $1/3$ of the work?

Answer: 4

Solution: Time for $1/3$ of the work = $12 \times (1/3) = 4$ days.

[GATE-style]

Q3. [MCQ · 2 marks]

A and B together can complete a work in 6 days. A alone can complete it in 10 days. In how many days can B alone complete the work?

- (A) 15
- (B) 12
- (C) 20
- (D) 18

Answer: (A)

Solution: Combined rate = $1/6$. A's rate = $1/10$. B's rate = $1/6 - 1/10 = 5/30 - 3/30 = 2/30 = 1/15$. B's time = 15 days.

[GATE-style]

Q4. [NAT · 2 marks]

A, B, and C can complete a work in 6, 8, and 12 days respectively. If they all work together, how many days will they take to complete the work (correct to 2 decimals)?

Answer: 2.67

Solution: Combined rate = $1/6 + 1/8 + 1/12 = 4/24 + 3/24 + 2/24 = 9/24 = 3/8$. Time = $8/3 \approx 2.67$ days.

[GATE-style]

Q5. [MCQ · 1 mark]

A can do a work in 20 days. He works for 5 days and then leaves. What fraction of the work is left?

- (A) $3/4$
- (B) $1/4$
- (C) $1/2$
- (D) $2/3$

Answer: (A)

Solution: Work done = $5/20 = 1/4$. Remaining work = $1 - 1/4 = 3/4$.

[GATE-style]

Q6. [NAT · 2 marks]

A does a work in 15 days and B does it in 10 days. They work together for 3 days, and then A leaves. In how many more days will B alone finish the remaining work?

Answer: 5

Solution: Combined rate = $1/15 + 1/10 = 2/30 + 3/30 = 5/30 = 1/6$. Work done in 3 days = $3 \times 1/6 = 1/2$. Remaining work = $1/2$. B's rate = $1/10$, so B needs $(1/2)/(1/10) = 5$ more days.

[GATE-style]

Q7. [MCQ · 1 mark]

If 6 men can complete a work in 12 days, how many days will 8 men take to complete the same work (assuming equal individual work rates)?

- (A) 9
- (B) 10
- (C) 8
- (D) 12

Answer: (A)

Solution: Total man-days = $6 \times 12 = 72$. Days needed for 8 men = $72/8 = 9$.

[GATE-style]

Q8. [MSQ · 2 marks]

Which of the following statements about time-and-work problems are TRUE?

- (A) Work rate is generally defined as $(1 / \text{time taken to complete the work})$
- (B) The combined work rate of multiple people working together is the sum of their individual work rates
- (C) If a person is twice as efficient as another, they take twice as long to complete the same work
- (D) Total work can conveniently be represented as 1 unit (or as the LCM of the individual completion times)

Answer: (A), (B), (D)

Solution: (C) is false -- being MORE efficient means taking LESS time, not more; a person twice as efficient takes HALF the time, not twice.

[GATE-style]

Q9. [NAT · 2 marks]

A pipe can fill a tank in 8 hours. Another pipe can empty the full tank in 12 hours. If both pipes are opened together on an empty tank, how many hours will it take to fill the tank?

Answer: 24

Solution: Fill rate = $1/8$, drain rate = $1/12$. Net rate = $1/8 - 1/12 = 3/24 - 2/24 = 1/24$. Time to fill = 24 hours.

[GATE-style]

Q10. [MCQ · 1 mark]

A is thrice as efficient as B. If B alone can complete a work in 30 days, in how many days can A alone complete it?

- (A) 10
- (B) 15

(C) 20

(D) 90

Answer: (A)

Solution: A's time = B's time / 3 = $30/3 = 10$ days.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

A can complete a work in 18 days, and B can complete it in 9 days. Working together, how many days will they take?

Answer: 6

Solution: Combined rate = $1/18 + 1/9 = 1/18 + 2/18 = 3/18 = 1/6$. Time = 6 days.

[New Practice]

Q12. [NAT · 1 mark]

A can do a work in 24 days. How many days will A take to complete $3/4$ of the work?

Answer: 18

Solution: Time = $24 \times (3/4) = 18$ days.

[New Practice]

Q13. [MCQ · 2 marks]

A and B together can finish a work in 12 days. B alone can finish it in 20 days. In how many days can A alone finish the work?

(A) 30

(B) 25

(C) 24

(D) 18

Answer: (A)

Solution: Combined rate = $1/12$. B's rate = $1/20$. A's rate = $1/12 - 1/20 = 5/60 - 3/60 = 2/60 = 1/30$. A's time = 30 days.

[New Practice]

Q14. [NAT · 2 marks]

A, B, and C can do a work in 10, 15, and 30 days respectively. Working together, how many days will they take?

Answer: 5

Solution: Combined rate = $1/10 + 1/15 + 1/30 = 3/30 + 2/30 + 1/30 = 6/30 = 1/5$. Time = 5 days.

[New Practice]

Q15. [MCQ · 1 mark]

A can complete a job in 16 days. After working for 4 days, what fraction of the work remains?

(A) $3/4$

(B) $1/2$

(C) $2/3$

(D) $1/4$

Answer: (A)

Solution: Work done = $4/16 = 1/4$. Remaining = $1 - 1/4 = 3/4$.

[New Practice]

Q16. [NAT · 2 marks]

A does a work in 20 days, B does it in 30 days. They work together for 5 days, then A leaves. How many more days will B need to finish the remaining work (correct to 2 decimals)?

Answer: 17.5

Solution: Combined rate = $1/20 + 1/30 = 3/60 + 2/60 = 5/60 = 1/12$. Work done in 5 days = $5/12$. Remaining = $7/12$. B's rate = $1/30$, so days needed = $(7/12)/(1/30) = 210/12 = 17.5$.

[New Practice]

Q17. [MCQ · 1 mark]

If 10 men can complete a work in 15 days, how many days will 6 men take to complete the same work?

- (A) 25
- (B) 20
- (C) 18
- (D) 30

Answer: (A)

Solution: Total man-days = $10 \times 15 = 150$. Days for 6 men = $150/6 = 25$.

[New Practice]

Q18. [MSQ · 2 marks]

Which of the following are TRUE regarding time-and-work problems involving pipes/taps?

- (A) An inlet (filling) pipe contributes a positive rate; an outlet (draining) pipe contributes a negative rate
- (B) If the net rate of all pipes combined is zero, the tank's water level stays constant (assuming it starts neither empty nor overflowing)
- (C) Two pipes can never be used together in the same problem
- (D) The combined rate of multiple pipes is the algebraic sum of their individual rates (accounting for sign)

Answer: (A), (B), (D)

[New Practice]

Q19. [NAT · 2 marks]

A pipe fills a tank in 6 hours. A drainage pipe can empty the same full tank in 10 hours. If both are opened together on an empty tank, how many hours will it take to fill the tank?

Answer: 15

Solution: Fill rate = $1/6$, drain rate = $1/10$. Net rate = $1/6 - 1/10 = 5/30 - 3/30 = 2/30 = 1/15$. Time = 15 hours.

[New Practice]

Q20. [MCQ · 2 marks]

A is twice as efficient as B. If A and B together can complete a work in 12 days, how many days would A alone take to complete the work?

- (A) 18
- (B) 24
- (C) 36
- (D) 15

Answer: (A)

Solution: Let B's rate = r , so A's rate = $2r$. Combined rate = $3r = 1/12 \Rightarrow r = 1/36$. A's rate = $2/36 = 1/18$. A's time = 18 days.

Topic 5: Nested Loop Complexity

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [MCQ · 1 mark]

For two nested loops, both running from 1 to n (outer and inner each n iterations, independent), the total time complexity is:

- (A) $O(n)$
- (B) $O(n \log n)$
- (C) $O(n^2)$
- (D) $O(2n)$

Answer: (C)

[GATE-style]

Q2. [NAT · 2 marks]

For nested loops where the outer loop runs n times and, for each outer iteration i , the inner loop runs i times (i.e. for($i=1$ to n) for($j=1$ to i)), the total number of inner-loop iterations is $n(n+1)/2$. Compute this for $n=10$.

Answer: 55

Solution: $n(n+1)/2 = 10 \times 11/2 = 55$.

[GATE-style]

Q3. [MCQ · 2 marks]

The nested-loop pattern in the previous question (inner loop runs i times when the outer loop variable is i) has overall time complexity:

- (A) $O(n)$
- (B) $O(n \log n)$
- (C) $O(n^2)$
- (D) $O(n^3)$

Answer: (C)

Solution: $\sum_{i=1}^n i = n(n+1)/2 = \Theta(n^2)$.

[GATE-style]

Q4. [MCQ · 1 mark]

For three nested loops, each running independently from 1 to n , the total time complexity is:

- (A) $O(n)$
- (B) $O(n^2)$
- (C) $O(n^3)$
- (D) $O(3n)$

Answer: (C)

[GATE-style]

Q5. [NAT · 1 mark]

For a nested loop structure where the outer loop runs from $i=1$ to n , and for each outer iteration the inner loop runs from $j=1$ to n with $O(1)$ work per inner iteration, the total time is $\Theta(n^k)$. Find k .

Answer: 2

[GATE-style]

Q6. [MCQ · 2 marks]

Consider: `for(i=1;i<=n;i++) for(j=i;j<=n;j++) {...0(1)...}`. What is the time complexity?

- (A) $O(n)$
- (B) $O(n \log n)$
- (C) $O(n^2)$
- (D) $O(n^3)$

Answer: (C)

Solution: For each i , the inner loop runs $(n-i+1)$ times. Total = $\sum (n-i+1)$ for $i=1$ to $n = n(n+1)/2 = \Theta(n^2)$.

[GATE-style]

Q7. [MCQ · 2 marks]

Consider: `for(i=1;i<=n;i=i*2) for(j=1;j<=n;j++) {...0(1)...}`. What is the time complexity?

- (A) $O(n)$
- (B) $O(n \log n)$
- (C) $O(\log n)$
- (D) $O(n^2)$

Answer: (B)

Solution: The outer loop runs $O(\log n)$ times (doubling i each time); the inner loop runs $O(n)$ times per outer iteration $\Rightarrow O(n \log n)$ total.

[GATE-style]

Q8. [MCQ · 2 marks]

Consider: `for(i=1;i<=n;i++) for(j=1;j<=n;j=j*2) {...0(1)...}`. What is the time complexity?

- (A) $O(n)$
- (B) $O(n \log n)$
- (C) $O(\log n)$
- (D) $O((\log n)^2)$

Answer: (B)

Solution: The outer loop runs $O(n)$ times, and the inner loop runs $O(\log n)$ times per outer iteration $\Rightarrow O(n \log n)$.

[GATE-style]

Q9. [NAT · 2 marks]

Consider: `for(i=1;i<=n;i=i*2) for(j=1;j<=n;j=j*2) {...0(1)...}`. The time complexity is $\Theta((\log n)^k)$. Find k .

Answer: 2

Solution: Both loops independently run $O(\log n)$ times (each doubling its index), giving $O((\log n)^2)$ overall.

[GATE-style]

Q10. [MSQ · 2 marks]

Which of the following nested-loop patterns have overall time complexity $O(n^2)$?

- (A) for(i=1 to n) for(j=1 to n) {O(1)}
- (B) for(i=1 to n) for(j=1 to i) {O(1)}
- (C) for(i=1 to n) for(j=1 to 5) {O(1)} (inner loop bounded by the fixed constant 5, independent of n)
- (D) for(i=1 to n) for(j=i to n) {O(1)}

Answer: (A), (B), (D)

Solution: (C) has an inner loop bounded by a constant (5), independent of n, so the total work is $O(5n) = O(n)$, NOT $O(n^2)$.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

For for(i=1 to n) for(j=1 to n/2) {O(1)}, the total complexity is $\Theta(n^k)$. Find k.

Answer: 2

[New Practice]

Q12. [MCQ · 1 mark]

For for(i=1 to n) for(j=1 to 100) {O(1)} (inner loop bounded by a fixed constant 100), the time complexity is:

- (A) $O(n)$
- (B) $O(n^2)$
- (C) $O(\log n)$
- (D) $O(1)$

Answer: (A)

[New Practice]

Q13. [NAT · 2 marks]

For for(i=1 to n) for(j=1 to i*i) {O(1)}, the total number of iterations is $\sum_{i=1}^n i^2$. Compute this exactly for $n=3$.

Answer: 14

Solution: $1^2+2^2+3^2 = 1+4+9 = 14$ (matches the formula $n(n+1)(2n+1)/6 = 3 \times 4 \times 7/6 = 14$).

[New Practice]

Q14. [MCQ · 2 marks]

The nested loop in the previous question has overall asymptotic time complexity:

- (A) $O(n)$
- (B) $O(n^2)$
- (C) $O(n^3)$
- (D) $O(n^4)$

Answer: (C)

Solution: $\sum i^2 = \Theta(n^3)$ (the standard sum-of-squares formula grows cubically).

[New Practice]

Q15. [MSQ · 2 marks]

Which of the following statements about nested loops are TRUE?

- (A) Two independent nested loops each running n times give $O(n^2)$ total

(B) A loop nested inside another always multiplies their complexities together, regardless of structure

(C) If the inner loop's iteration count depends on the outer loop variable (e.g. triangular loops), the total must be computed via summation, not simple multiplication

(D) Nested loops always have worse complexity than a single loop

Answer: (A), (C)

Solution: (B) is false in the triangular case -- you cannot simply multiply $n \times n$ when the inner bound depends on i ; you must sum. (D) is false -- this is not a universal rule.

[New Practice]

Q16. [NAT · 1 mark]

For three independent nested loops, each running from 1 to n , the complexity is $\Theta(n^k)$. Find k .

Answer: 3

[New Practice]

Q17. [MCQ · 2 marks]

Consider: `for(i=n; i>=1; i=i/2) {for(j=1; j<=n; j++){O(1)}}`. What is the time complexity?

(A) $O(n)$

(B) $O(n \log n)$

(C) $O(\log n)$

(D) $O(n^2)$

Answer: (B)

Solution: The outer loop halves i each time, running $O(\log n)$ times; the inner loop runs $O(n)$ times per outer iteration $\Rightarrow O(n \log n)$ total.

[New Practice]

Q18. [NAT · 2 marks]

A triangular nested loop `for(i=1 to n) for(j=1 to n-i) {O(1)}` performs a total of $n(n-1)/2$ iterations. Compute this for $n=6$.

Answer: 15

Solution: $n(n-1)/2 = 6 \times 5/2 = 15$.

[New Practice]

Q19. [MCQ · 1 mark]

Which is the correct asymptotic complexity of a "triangular" double loop (inner loop bound depends linearly on the outer index, as in `for(i=1 to n) for(j=1 to i)`)?

(A) $O(n)$

(B) $O(n \log n)$

(C) $O(n^2)$

(D) $O(n^3)$

Answer: (C)

[New Practice]

Q20. [MCQ · 1 mark]

`for(i=1 to  $n^2$ ) {O(1)}` (a single loop running n^2 times) has the SAME asymptotic time complexity as which nested-loop pattern?

- (A) for($i=1$ to n) for($j=1$ to n) $\{O(1)\}$
- (B) for($i=1$ to n) $\{O(1)\}$
- (C) for($i=1$ to $\log n$) for($j=1$ to n) $\{O(1)\}$
- (D) for($i=1$ to n) for($j=1$ to $\log n$) $\{O(1)\}$

Answer: (A)

Topic 6: Master Theorem - 5 Recurrence Problems

This topic includes five fully worked Master Theorem problems, spread across the sections below: (1) $T(n)=2T(n/2)+n$ [Case 2, $\Theta(n \log n)$], (2) $T(n)=4T(n/2)+n$ [Case 1, $\Theta(n^2)$], (3) $T(n)=T(n/2)+1$ [Case 2, $\Theta(\log n)$], (4) $T(n)=3T(n/2)+n^2$ [Case 3, $\Theta(n^2)$], and (5) $T(n)=2T(n/4)+n^{0.51}$ [Case 3, $\Theta(n^{0.51})$]. Since none of these are GATE DA/CSE verbatim past questions, all are labelled [GATE-style], written in the authentic Master-theorem-application format GATE uses. Several additional Master Theorem recurrences appear as further practice in Section B.

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [MCQ · 1 mark]

The Master theorem applies to recurrences of the form:

- (A) $T(n) = T(n-1) + f(n)$
- (B) $T(n) = aT(n/b) + f(n)$, for constants $a \geq 1$, $b > 1$
- (C) $T(n) = T(n/2) \times T(n/2)$
- (D) $T(n) = a \cdot f(n)$

Answer: (B)

[GATE-style, Master Theorem Problem 1: $T(n)=2T(n/2)+n$]

Q2. [MCQ · 2 marks]

For $T(n) = 2T(n/2) + n$, compare $f(n)=n$ with $n^{\log_b a} = n^{\log_2 2} = n^1$. Since $f(n) = \Theta(n^1)$, which Master theorem case applies, and what is $T(n)$?

- (A) Case 1: $T(n) = \Theta(n)$
- (B) Case 2: $T(n) = \Theta(n \log n)$
- (C) Case 3: $T(n) = \Theta(n)$
- (D) The Master theorem does not apply

Answer: (B)

[GATE-style, Master Theorem Problem 2: $T(n)=4T(n/2)+n$]

Q3. [MCQ · 2 marks]

For $T(n) = 4T(n/2) + n$, here $a=4$, $b=2$, so $n^{\log_b a} = n^{\log_2 4} = n^2$. Since $f(n)=n=O(n^{2-\epsilon})$ for $\epsilon=1$, which case applies, and what is $T(n)$?

- (A) Case 1: $T(n) = \Theta(n^2)$
- (B) Case 2: $T(n) = \Theta(n^2 \log n)$
- (C) Case 3: $T(n) = \Theta(n)$
- (D) The Master theorem does not apply

Answer: (A)

[GATE-style, Master Theorem Problem 3: $T(n)=T(n/2)+1$]

Q4. [NAT · 2 marks]

For $T(n) = T(n/2) + 1$, here $a=1$, $b=2$, so $n^{\log_b a} = n^0 = 1$. Since $f(n)=1=\Theta(n^0)$, Case 2 gives $T(n) = \Theta(\log n)$. Using $T(1)=1$, compute $T(32)$ by direct unrolling, and verify it matches $\log_2(32)+1$.

Answer: 6

Solution: $T(1)=1$, $T(2)=2$, $T(4)=3$, $T(8)=4$, $T(16)=5$, $T(32)=6$. Check: $\log_2(32)+1 = 5+1 = 6 \checkmark$.

[GATE-style, Master Theorem Problem 4: $T(n)=3T(n/2)+n^2$]

Q5. [MCQ · 2 marks]

For $T(n) = 3T(n/2) + n^2$, here $a=3$, $b=2$, so $n^{\log_b a} = n^{\log_2 3} \approx n^{1.585}$. Since $f(n)=n^2=\Omega(n^{1.585+\epsilon})$ and the regularity condition holds ($(3(n/2))^2=0.75n^2 \leq cn^2$ for $c=0.75<1$), which case applies, and what is $T(n)$?

- (A) Case 1: $T(n) = \Theta(n^{1.585})$
- (B) Case 2: $T(n) = \Theta(n^2 \log n)$
- (C) Case 3: $T(n) = \Theta(n^2)$
- (D) The Master theorem does not apply

Answer: (C)

[GATE-style, Master Theorem Problem 5: $T(n)=2T(n/4)+n^{0.51}$]

Q6. [MCQ · 2 marks]

For $T(n) = 2T(n/4) + n^{0.51}$, here $a=2$, $b=4$, so $n^{\log_b a} = n^{\log_4 2} = n^{0.5}$. Since $f(n)=n^{0.51}=\Omega(n^{0.5+\epsilon})$ for small ϵ , and the regularity condition holds, which case applies, and what is $T(n)$?

- (A) Case 1: $T(n) = \Theta(n^{0.5})$
- (B) Case 2: $T(n) = \Theta(n^{0.5} \log n)$
- (C) Case 3: $T(n) = \Theta(n^{0.51})$
- (D) The Master theorem does not apply

Answer: (C)

[GATE-style]

Q7. [MCQ · 1 mark]

In the Master theorem, "Case 1" (where $f(n)$ is polynomially smaller than $n^{\log_b a}$) results in a solution dominated by:

- (A) $f(n)$ itself
- (B) The recursive term $n^{\log_b a}$
- (C) $\log n$
- (D) A constant

Answer: (B)

[GATE-style]

Q8. [MCQ · 1 mark]

In the Master theorem, "Case 3" (where $f(n)$ is polynomially larger than $n^{\log_b a}$, and the regularity condition holds) results in a solution dominated by:

- (A) $f(n)$ itself
- (B) The recursive term $n^{\log_b a}$
- (C) $\log n$
- (D) A constant

Answer: (A)

[GATE-style]

Q9. [MCQ · 1 mark]

In the Master theorem, "Case 2" (where $f(n) = \Theta(n^{\log_b a})$) results in a solution of:

- (A) $\Theta(f(n))$

- (B) $\Theta(n^{\log_b a})$
- (C) $\Theta(n^{\log_b a} \cdot \log n)$
- (D) $\Theta(\log n)$

Answer: (C)

[GATE-style]

Q10. [MSQ · 2 marks]

Which of the following are required conditions/assumptions for applying the standard Master theorem to $T(n)=aT(n/b)+f(n)$?

- (A) $a \geq 1$ and $b > 1$ must be constants
- (B) $f(n)$ must be asymptotically positive
- (C) For Case 3, an additional "regularity condition" ($af(n/b) \leq cf(n)$ for some $c < 1$ and large n) must hold
- (D) a and b can be arbitrary functions of n

Answer: (A), (B), (C)

Solution: (D) is false -- a and b must be constants for the standard Master theorem to apply.

Section B — New Practice Questions (10)

[New Practice]

Q11. [MCQ · 2 marks]

For $T(n) = 8T(n/2) + n^2$, find $n^{\log_b a}$ and determine which Master theorem case applies.

- (A) Case 1, $T(n)=\Theta(n^3)$
- (B) Case 2, $T(n)=\Theta(n^3 \log n)$
- (C) Case 3, $T(n)=\Theta(n^2)$
- (D) The Master theorem does not apply

Answer: (A)

Solution: $a=8$, $b=2$, $\log_2 8=3$, so n^3 . $f(n)=n^2=O(n^{3-1}) \Rightarrow$ Case 1 $\Rightarrow T(n)=\Theta(n^3)$.

[New Practice]

Q12. [NAT · 1 mark]

Using $T(n)=8T(n/2)+n^2$, $T(1)=1$, compute $T(2)$ by direct unrolling.

Answer: 12

Solution: $T(1)=1$. $T(2)=8(1)+4=12$.

[New Practice]

Q13. [MCQ · 2 marks]

For $T(n) = T(n/3) + n$, find $n^{\log_b a}$ and determine the Master theorem case.

- (A) Case 1, $T(n)=\Theta(n^0)$
- (B) Case 2, $T(n)=\Theta(\log n)$
- (C) Case 3, $T(n)=\Theta(n)$
- (D) The Master theorem does not apply

Answer: (C)

Solution: $a=1$, $b=3$, $\log_3 1=0$, so $n^0=1$. $f(n)=n=\Omega(n^{0+1})$, and the regularity condition holds ($1 \times (n/3) \leq c \times n$ for $c=1/3 < 1$) $\Rightarrow$ Case 3 $\Rightarrow T(n)=\Theta(n)$.

[New Practice]

Q14. [MCQ · 2 marks]

For $T(n) = 5T(n/5) + n$, find $n^{\log_b a}$ and determine the case.

- (A) Case 1, $T(n) = \Theta(n)$
- (B) Case 2, $T(n) = \Theta(n \log n)$
- (C) Case 3, $T(n) = \Theta(n^2)$
- (D) The Master theorem does not apply

Answer: (B)

Solution: $a=5$, $b=5$, $\log_5 5=1$, so $n^1=n$. $f(n)=n=\Theta(n^1) \Rightarrow$ Case 2 $\Rightarrow T(n)=\Theta(n \log n)$.

[New Practice]

Q15. [NAT · 2 marks]

Using $T(n)=5T(n/5)+n$, $T(1)=1$, compute $T(5)$ by direct unrolling.

Answer: 10

Solution: $T(1)=1$. $T(5)=5(1)+5=10$.

[New Practice]

Q16. [MCQ · 2 marks]

For $T(n) = 9T(n/3) + n$, find $n^{\log_b a}$ and determine the case.

- (A) Case 1, $T(n) = \Theta(n^2)$
- (B) Case 2, $T(n) = \Theta(n^2 \log n)$
- (C) Case 3, $T(n) = \Theta(n)$
- (D) The Master theorem does not apply

Answer: (A)

Solution: $a=9$, $b=3$, $\log_3 9=2$, so n^2 . $f(n)=n=O(n^{2-1}) \Rightarrow$ Case 1 $\Rightarrow T(n)=\Theta(n^2)$.

[New Practice]

Q17. [NAT · 1 mark]

Using $T(n)=9T(n/3)+n$, $T(1)=1$, compute $T(3)$ by direct unrolling.

Answer: 12

Solution: $T(1)=1$. $T(3)=9(1)+3=12$.

[New Practice]

Q18. [MCQ · 2 marks]

For $T(n) = 2T(n/2) + n \log n$, is the Master theorem's standard three-case version directly applicable? (This is a classic edge case.)

- (A) Yes, it falls cleanly into Case 2 since $n \log n = \Theta(n^1)$
- (B) No -- $n \log n$ is NOT polynomially larger than n (since $\log_b a=1$), so it falls into a gap not covered by the basic three cases (needing the extended Master theorem or another method); a common result is $T(n)=\Theta(n \log^2 n)$
- (C) Yes, it falls into Case 1
- (D) Yes, it falls into Case 3

Answer: (B)

Solution: $n \log n$ grows faster than n but not by a polynomial factor n^ϵ for any $\epsilon > 0$, so it lies in the gap between Case 1 and Case 2 not covered by the basic Master theorem.

[New Practice]

Q19. [MCQ · 2 marks]

For $T(n) = 4T(n/2) + n^3$, find $n^{\log_b a}$ and determine the case.

- (A) Case 1, $T(n) = \Theta(n^2)$
- (B) Case 2, $T(n) = \Theta(n^2 \log n)$
- (C) Case 3, $T(n) = \Theta(n^3)$
- (D) The Master theorem does not apply

Answer: (C)

Solution: $a=4$, $b=2$, $\log_2 4=2$, so n^2 . $f(n)=n^3=\Omega(n^{2+1})$, with regularity $4(n/2)^3=0.5n^3 \leq cn^3$ for $c=0.5<1$
 $\Rightarrow$ Case 3 $\Rightarrow T(n)=\Theta(n^3)$.

[New Practice]

Q20. [NAT · 2 marks]

Using $T(n)=4T(n/2)+n^3$, $T(1)=1$, compute $T(2)$ by direct unrolling.

Answer: 12

Solution: $T(1)=1$. $T(2)=4(1)+8=12$.

Topic 7: Uninformed Search - Uniform Cost Search (UCS)

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [MCQ · 1 mark]

Uniform Cost Search (UCS) expands nodes in order of:

- (A) Increasing depth (number of edges) from the start, like BFS
- (B) Increasing cumulative path cost $g(n)$ from the start node
- (C) Decreasing heuristic value
- (D) Random order

Answer: (B)

[GATE-style]

Q2. [MCQ · 1 mark]

UCS is most closely related to (and reduces to, on an explicit graph) which classical shortest-path algorithm?

- (A) Bellman-Ford
- (B) Dijkstra's algorithm
- (C) Floyd-Warshall
- (D) Kruskal's algorithm

Answer: (B)

[GATE-style]

Q3. [MCQ · 1 mark]

UCS uses which data structure to manage its frontier?

- (A) A simple FIFO queue
- (B) A stack (LIFO)
- (C) A priority queue ordered by path cost $g(n)$
- (D) A hash table only

Answer: (C)

[GATE-style]

Q4. [MCQ · 1 mark]

UCS reduces to plain BFS when:

- (A) All edge costs are negative
- (B) All edge (step) costs are equal (e.g. all = 1)
- (C) The graph has no edges
- (D) The heuristic used is admissible

Answer: (B)

[GATE-style]

Q5. [MSQ · 2 marks]

Which of the following are properties of Uniform Cost Search?

- (A) It is complete, provided the branching factor is finite and every edge cost is bounded below by some positive constant $\epsilon > 0$
- (B) It is optimal (finds the minimum-cost path to the goal), provided all edge costs are non-negative
- (C) It requires a heuristic function to operate
- (D) The goal test is applied when a node is selected for expansion (removed from the frontier), not when it is first generated

Answer: (A), (B), (D)

Solution: (C) is false -- UCS is an UNINFORMED search algorithm; it uses no heuristic function at all (that is precisely what distinguishes it from A*).

[GATE-style]

Q6. [NAT · 2 marks]

A graph has start node S and edges: S→A (cost 1), S→B (cost 4), A→B (cost 1), A→C (cost 5), B→C (cost 1). Find the minimum cost to reach node C from S.

Answer: 3

Solution: Paths to C: S-A-C: 1+5=6. S-A-B-C: 1+1+1=3. S-B-C: 4+1=5. Minimum cost = 3, via S-A-B-C.

[GATE-style]

Q7. [MCQ · 1 mark]

Using the graph from the previous question, which path does UCS return as the optimal path to C?

- (A) S-A-C
- (B) S-B-C
- (C) S-A-B-C
- (D) S-C directly

Answer: (C)

[GATE-style]

Q8. [MCQ · 1 mark]

If UCS finds that a node already in the frontier can be reached via a new path with LOWER cumulative cost than its currently recorded cost, UCS should:

- (A) Ignore the new path entirely
- (B) Update (relax) the node's cost and parent pointer to reflect the cheaper path found
- (C) Immediately terminate the search
- (D) Restart the search from scratch

Answer: (B)

[GATE-style]

Q9. [MCQ · 1 mark]

Compared to plain BFS, UCS is necessary (rather than BFS) whenever:

- (A) All step costs are identical
- (B) Step costs vary (are non-uniform) across edges, and the least-cost path (not necessarily the fewest-edges path) is required
- (C) The search space is a tree
- (D) Memory is extremely limited

Answer: (B)

[GATE-style]

Q10. [MCQ · 2 marks]

The time and space complexity of UCS, in the worst case, is typically expressed as:

- (A) $O(b^d)$, where d is the depth of the shallowest goal, as in BFS
- (B) $O(b^{1+\lceil C^*/\epsilon \rceil})$, where C^* is the cost of the optimal solution and ϵ is the smallest edge cost
- (C) $O(\log n)$
- (D) $O(n)$, always

Answer: (B)

Solution: Since UCS explores nodes in order of increasing path cost rather than increasing depth, its complexity is naturally expressed in terms of the optimal solution cost C^* and the smallest edge cost ϵ , which can correspond to a much larger effective "depth" than a simple BFS bound.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 2 marks]

A graph has start S and edges $S \rightarrow A$ (2), $S \rightarrow B$ (5), $A \rightarrow B$ (1), $A \rightarrow D$ (7), $B \rightarrow D$ (2). Find the minimum cost from S to D .

Answer: 5

Solution: Paths: $S-A-D$: $2+7=9$. $S-A-B-D$: $2+1+2=5$. $S-B-D$: $5+2=7$. Minimum = 5, via $S-A-B-D$.

[New Practice]

Q12. [MCQ · 1 mark]

For the graph above, the optimal path UCS returns for reaching D is:

- (A) $S-A-D$
- (B) $S-B-D$
- (C) $S-A-B-D$
- (D) None reach D

Answer: (C)

[New Practice]

Q13. [MCQ · 1 mark]

In UCS, when a node is popped from the priority queue (frontier) for expansion and it turns out to be the goal, what should be done?

- (A) Immediately return it as the optimal solution, since UCS always expands nodes in non-decreasing order of path cost
- (B) Ignore it and continue searching
- (C) Restart UCS from that node
- (D) Discard it, since goals cannot be expanded

Answer: (A)

[New Practice]

Q14. [MSQ · 2 marks]

Which of the following distinguish UCS from Dijkstra's algorithm as typically presented?

- (A) UCS is usually described for finding the shortest path to a SINGLE specified goal node, stopping as soon as that goal is expanded

- (B) Dijkstra's algorithm, as classically presented, finds shortest paths from the source to ALL other nodes
- (C) The underlying mechanism (a priority queue ordered by cumulative cost) is essentially the same in both
- (D) UCS requires the graph to be fully known/explicit in advance, unlike Dijkstra's algorithm

Answer: (A), (B), (C)

Solution: (D) is false -- if anything the emphasis is reversed: UCS is often applied to implicit search spaces generated on the fly, while Dijkstra's algorithm is classically presented on a fully explicit graph.

[New Practice]

Q15. [NAT · 1 mark]

UCS runs on a graph where the smallest edge cost is $\epsilon=1$ and the optimal solution cost is $C^*=6$. Using the complexity estimate exponent $(1+\lceil C^*/\epsilon \rceil)$, compute this exponent.

Answer: 7

Solution: $1+\lceil 6/1 \rceil = 1+6 = 7$.

[New Practice]

Q16. [MCQ · 1 mark]

Which of the following is TRUE about UCS's optimality guarantee?

- (A) UCS is optimal even if some edge costs are negative
- (B) UCS is optimal only when all edge costs are non-negative
- (C) UCS is never optimal
- (D) UCS's optimality depends on having an admissible heuristic

Answer: (B)

Solution: With negative edge costs, a node could later be reached via an even lower-cost path after already being expanded, breaking UCS's core assumption; (D) is also false since UCS uses no heuristic at all.

[New Practice]

Q17. [MCQ · 1 mark]

Which algorithm would you choose to guarantee the cheapest path in a graph with varying positive edge weights, using NO heuristic information at all?

- (A) DFS
- (B) Plain BFS
- (C) Uniform Cost Search (UCS)
- (D) Simple hill climbing

Answer: (C)

[New Practice]

Q18. [NAT · 2 marks]

A graph has start S and edges $S \rightarrow A$ (3), $S \rightarrow B$ (1), $A \rightarrow G$ (1), $B \rightarrow G$ (5). Find the minimum cost from S to goal G.

Answer: 4

Solution: Paths: S-A-G: $3+1=4$. S-B-G: $1+5=6$. Minimum = 4, via S-A-G.

[New Practice]

Q19. [MCQ · 1 mark]

For the graph in the previous question, which path does UCS return?

- (A) S-A-G
- (B) S-B-G
- (C) Both equally
- (D) Neither

Answer: (A)

[New Practice]

Q20. [MCQ · 1 mark]

In UCS, the frontier is best described as a set of nodes that have been:

- (A) Fully expanded and never revisited
- (B) Generated (discovered) but not yet expanded, ordered by cumulative path cost
- (C) Permanently discarded
- (D) Selected as part of the final solution path

Answer: (B)
